

**RAJIV GANDHI PROUDYOGIKI VISHWAVIDYALAYA, BHOPAL**

**New Scheme Based On AICTE Flexible Curricula**

**CSE Internet of Things and Cyber Security Including Block Chain Technology, V-Semester**

**IS 501- FOUNDATIONS OF BLOCKCHAIN TECHNOLOGY**

**UNIT-I**

Introduction to Blockchain: Need for Distributed Record Keeping, Blockchain architecture, blockheader detailed design, Abstract Models for Blockchain, Proof of Work (PoW), liveness and fairness, Proof of Stake (PoS) based Chains, Hybrid models (PoW + PoS); Types of Blockchain..

**UNIT-II**

Blockchain Consensus Algorithm challenges and solutions, Modeling faults and adversaries, Byzantine Models of Fault tolerance; Zero Knowledge proofs and protocols in Blockchain.

**UNIT-III**

Introduction to cryptographic basics for cryptocurrency - a short description of Hashing, digital signature schemes, encryption schemes and elliptic curve cryptography, verifiable random functions.

**UNIT-IV**

Blockchain 2.0: Introduction to Ethereum, Ethereum Virtual Machine (EVM), Wallets for Ethereum, Solidity, Smart Contracts, Attacks on smart contracts, The Turing Completeness of Smart Contract Languages and verification challenges. Blockchain 3.0: Hyperledger implementation on Ethereum, the plug and play platform and mechanisms in permissioned blockchain.

**UNIT-V**

Application of Blockchain: Bitcoin- Bitcoin consensus, Wallet, Bitcoin Blocks, Merkle Tree, hardness of mining, transaction verifiability, anonymity, forks, double spending, mathematical analysis of properties of Bitcoin. Altcoins. Medical record management systems, DNS records.

**BOOKS RECOMMENDED:**

1. Josh Thompson, 'Blockchain: The Blockchain for Beginnings, Guide to Blockchain Technology and Blockchain Programming', Create Space Independent Publishing Platform, 2017.
2. Arvind Narayanan, Joseph Bonneau, Edward Felten, Andrew Miller, and Steven Goldfeder, 'Bitcoin and cryptocurrency technologies: a comprehensive introduction', Princeton University Press, 2016.
3. Kumar Saurabh, Ashutosh Saxena, 'Blockchain Technology: Concepts and Applications', Wiley, 2020.